

Research Methodology

Introduction to Research

Equipping You with Research Depth and
Industry Skills

By:

Dr. Zohair Ahmed



 www.youtube.com/@ZohairAI 

 www.begindiscovery.com

Thinking Like a Researcher

- Before diving into the course content, take a moment to reflect:
 - **What is something you've been genuinely curious about recently - in your studies, work, or everyday life?**
- Write a brief response (1–2 sentences).
- There is no need to refine or polish it, just try to capture an idea that genuinely interests you.
- Think of it as planting the seed of a research question.

What is Research

- **Multiple definitions exist** – no single universal answer
- Common observation identified in interviews:
 - Not just *reading* or *collecting info*
 - Must **add something original** (even small, specific)
 - Contribution should be **meaningful to the body of knowledge**

Why Do Research

- **Personal Reasons**

- Curiosity & intellectual growth
- Developing critical thinking & problem-solving skills

- **Academic/Scientific Reasons**

- Expand knowledge in a field
- Challenge or refine existing theories

- **Social/Practical Reasons**

- Provide solutions to real-world problems
- Influence policy, industry, or community



Why Do Research

- **Relevance:** Must matter to society, academia, or practice
- **Systematic Process:** Clear methods & transparent reasoning
- **Balance:** Feasible scope **vs. meaningful impact**
- **Finite End:** Every project must have a conclusion point

Research in Practice

- Involves multiple activities:
 - **Thinking time** (conceptual exploration)
 - **Reading** (literature review)
 - **Writing** (drafting, reflection, structuring ideas)
 - **Discussion** (feedback, peer review)
- Combined, these form the **research method**
- **Example for AI Students:** Reading AI ethics papers, brainstorming gaps, writing a proposal on bias detection.



Scope of a Research Question

- **Too Broad:** “How does AI affect society?” → impossible to research fully.
- **Too Narrow:** “How does AI affect my neighbor’s online shopping?” → insignificant & non-generalizable.
- **Balanced:** “What are the impacts of AI-based hiring tools on recruitment fairness in developing countries?”



What Makes Good Research?

- Adds *new knowledge* or *new perspective*
- **Systematic**: replicable, well-structured methods
- **Relevant**: important for academia, society, or industry
- **Ethical**: respects participants, avoids plagiarism, honest reporting
- **Dynamic**: adapts as new insights arise

Conversation with Prof. Sandra Halperin

- Author of *Political Research Methods and Practical Skills*
- Topic: **What is Research?** and **What Makes a Good Research Question?**
- Based on her experience as a **student** and a **professional academic**
-  Watch Prof. Sandra Halperin explain what makes a good research question (Video Link)

What is Research?

- **Definition:** *Systematic inquiry* that helps us:
 - Make sense of the world
 - Interpret debates and contemporary issues
- **Key point:** Research ≠ just reading books or collecting facts
- **Purpose:** Orient thinking, define relevance, and guide choices



Role of Research Questions

- A **research question** is the *starting point* of research
- Helps to:
 - Orient everything you do
 - Decide what's relevant vs. irrelevant
 - Determine what to include/exclude
 - Guide allocation of time/space in writing
- **Example:** In AI research → “How to reduce hallucinations in LLMs?” guides dataset, methods, evaluation.

Attributes of a Good Research Question

- **1. Significance**
- Must matter to a broad community of scholars
- Addresses an important issue/problem/puzzle
- Rooted in **real-world problems** or **theoretical frameworks**
- We should *care* about it



Attributes of a Good Research Question

- **2. Researchable**
- Must be **answerable through systematic research**
- Avoid:
 - Metaphysical or overly philosophical questions
 - Questions too broad to study effectively
- Requires clear **data sources and methods**

Attributes of a Good Research Question

- **3. Novelty / Unsettled**
- Should not be **definitively answered** already
- Must allow for **new knowledge contribution**:
 - Strengthen existing answers
 - Apply to new contexts or cases
 - Challenge or refine accepted views
- **Not** just summarizing others' views → that's an essay, not research



Putting it Together

- **A Good Research Question is:**
- **Significant** – it matters
- **Researchable** – it's answerable with systematic inquiry
- **Unsettled** – it leaves room for new contributions
- **Example:**
- Weak: “Does AI exist?” (too broad, partly metaphysical)
- Strong: “What factors affect the accuracy of AI models in detecting hate speech across languages?”



What is a Research Question?

- A **research question** is the **core question** that guides your entire study.

It tells you:

- **What you want to know** (the focus).
- **Why it matters** (the significance).
- **How you will investigate it** (the method).
- Without a research question, research becomes just *collecting random information*.

Why do we need a research question? Can't I just collect data, run experiments, improve a model, and publish?



Why You Need a Research Question

- **1. Gives Direction & Focus**
- Without a research question, your work is like **boating without a compass**.
- You may collect data and run experiments, but:
 - What exactly are you testing?
 - Why this dataset/model, not another?
 - How will you know if your result is meaningful?
- A research question focuses your study so every step has a *purpose*.

Why You Need a Research Question

- 2. Turns Work Into “Research,” Not Just Engineering
- **Engineering/implementation:** “I improved accuracy of Model X by 2%.”
- **Research:** “*How does adding attention improve accuracy in humor detection in Arabic compared to baseline CNN models?*”
- The first is a **result**, but the second is a **contribution** to knowledge and that’s what journals/conferences care about.

Why You Need a Research Question

- **Helps Identify the Contribution**
- A research question makes it clear:
 - What gap are you addressing?
 - What new knowledge are you adding to the field?
- Reviewers/editors ask: *“What question does this paper answer?”*
 - If you can’t state it, the work looks like a **technical report** instead of a **research paper**.



Why You Need a Research Question

- **Avoids Wasted Effort**
- Without a clear question, you might:
 - Collect unnecessary data
 - Try random experiments
 - End up with results that are interesting but don't fit into a publishable narrative.
- A research question makes sure you design **relevant experiments** that contribute to your field.



Why You Need a Research Question

- **Example in AI**
- Without Research Question:

“I trained a transformer on Arabic jokes and improved accuracy by 3%.”

 - OK, but so what? Why does it matter?
- With Research Question:

“Does using transformer-based embeddings significantly improve Arabic humor detection compared to RNN-based embeddings?”

 - Clear goal → Collect relevant data → Run focused experiments → Answer systematically.
 - Contribution is clear: you’re testing a **specific assumption** and adding to the body of knowledge.

Small Group Activity

- Form groups of **3 to 4 students**.
- Your teacher will assign you a **broad topic**:
 - AI in healthcare
 - AI in social media
 - AI in cybersecurity
 - AI in education
- In your group, **write two research questions**:
 - **One “bad” question** (too broad, too narrow, trivial, or unanswerable).
 - **One “good” question** (specific, researchable, significant, unsettled).
- Be ready to share your answers with the class.
- It Should be:
 - **Significant** (matters to society/field)
 - **Researchable** (can collect/analyze data)
 - **Unsettled** (not fully answered yet)

Examples

- **Topic: AI in Healthcare**
- Bad: *“Is AI useful in healthcare?”*
 - Too broad, vague, subjective.
- Improved: *“How accurate are AI-based diagnostic tools in detecting early-stage liver tumors compared to human radiologists?”*
 - Clear, Researchable, Significant.
- **Topic: AI in Social Media**
- Bad: *“Is AI good or bad for social media?”*
 - Value judgment, no measurable outcome.
- Improved: *“How effective are AI algorithms in detecting cyberbullying on Arabic-language social media platforms?”*
 - Narrow focus, Real data possible, Socially important.



Examples

- **Topic: AI in Cybersecurity**

- Bad: *“Will AI stop all hackers?”*

- Unrealistic, unanswerable, absolute.

- Improved: *“What are the strengths and weaknesses of AI-based intrusion detection systems compared to traditional rule-based systems in corporate networks?”*

- Feasible, Comparative, Novel.

- **Topic: AI in Education**

- Bad: *“Is AI helpful for students?”*

- Too vague, trivial.

- Improved: *“What impact do AI-powered tutoring chatbots have on postgraduate students’ writing skills compared to traditional peer feedback?”*

- Specific, Researchable, Relevant to your students.



Validation

- How Validate:
- **Clarity** → Is it a real question or just a broad topic?
- **Significance** → Why does it matter? Who benefits?
- **Researchability** → Can it be answered with available data/methods?
- **Novelty** → Is there something new (gap, application, or context)?
- **Scope** → Manageable within a semester? Not too broad or trivial?



Do we find the gap first, or the question first?



Path 1: Start with the Gap

- **Path 1: Start with the Gap → Then Frame Question**
- You do a **literature review** → read papers → notice something missing.
- That “gap” sparks a research question.
- Example in AI:
- You review papers on **hate speech detection**.
- Gap: Most focus on English, very few on Arabic.
- Question: *“How effective are transformer-based models in detecting Arabic hate speech compared to English?”*



Path 2: Start with the Question

- **Path 2: Start with the Question → Then Identify Gap**
- You have a curiosity/idea first.
- Then you check the literature to see:
 - Has it been answered already?
 - If yes → refine it.
 - If no → that's your gap.
- Example in AI:
- Curiosity: *“Can ChatGPT help students write better research proposals?”*
- You check the literature: Almost no studies in academic writing support for AI tools.
- Gap confirmed → research question is valid.



Which Is Better for Students?

- For **beginners**: starting with a **broad curiosity/question** is easier. Literature review then helps refine it into a gap.
- For **advanced researchers**: starting with a **literature gap** is common, because they already know the field deeply.
- Both paths converge to the same point:
A good research question that fills a meaningful gap.

Types of Research

- **1. Basic Research (Theory-driven)**
- **Goal:** Expand general knowledge.
- **Question Type:** *“Why does this happen?”*
- **Example (NLP):** *Studying how attention mechanisms in transformers represent long sequences.*

Overlap: May later become applied if someone uses it to improve translation models.



Types of Research

- **2. Applied Research (Problem-solving)**
- **Goal:** Solve a specific, practical problem.
- **Question Type:** *“How can we use this knowledge?”*
- **Example (NLP):** *Building an AI system to detect fake news in Arabic.*

Overlap: Uses **basic knowledge** (e.g., embeddings) to solve a real-world problem.



Types of Research

- **3. Exploratory Research (Early-stage discovery)**
- **Goal:** Investigate a new/unclear area.
- **Question Type:** *“What’s going on here? What questions should we ask?”*
- **Example (AI in education):** *Exploring how students use ChatGPT for assignments before we design formal studies.*

Overlap: Can lead to **basic** (new theories of learning) or **applied** (AI teaching tools).

- **In short:**
- **Exploratory = finding the question**
- **Basic = finding the explanation**
- **Applied = finding the solution**

Research Question Quiz

- 1. Which of the following is the most appropriate exploratory research question for this scenario?

A. Is social media harming young people?

B. How do students aged 14–18 describe the influence of social media on their study habits and focus?

C. How can schools stop students from using social media during study time?

D. Why do students use TikTok so much?

- A (too vague, value-laden)
- B (open, descriptive, exploratory)
- C (prescriptive, not exploratory)
- D (too broad, not study-focused)
- **Correct: B**



Research Question Quiz

- **2. Which of the following research questions is too broad or unfocused for a student research project?**
 - A. How do students perceive the benefits and drawbacks of using study apps that include social media features?
 - B. What is the psychological impact of all social media platforms on global education outcomes?
 - C. What strategies do students report using to manage distractions while studying?
 - D. How do high-achieving and low-achieving students describe their screen time routines during term-time?
- A (manageable)
- B (far too broad: “all platforms” + “global”)
- C (focused and researchable)
- D (manageable)
- **Correct: B**



Research Question Quiz

- **3. What is a key weakness in the question: “Should students be banned from using TikTok during the school week?”**
 - A. It is not specific to educational settings
 - B. It only applies to TikTok, not all social media
 - C. It is opinion-based and prescriptive, rather than exploratory or research-focused
 - D. It is a very popular topic
- A. It is, since it’s about school
- B. Focus is not the main issue
- C. Should students be banned is not a neutral research question
- D. Popularity is not the problem
- **Correct: C**



Research Question Quiz

- 4. Which of the following best explains why the research question “Do students who use Instagram study less because they are addicted?” is problematic from a social science research perspective?
 - A. It narrows the focus too much by selecting only one social media platform
 - B. It conflates correlation with causation and introduces a value-laden term
 - C. It fails to define "study less" in operational terms
 - D. It lacks a proposed method for data collection
- A. scope is fine if justified
- B. assumes addiction → causality; “addicted” is judgmental
- C. true but secondary
- D. methods come later, not in the question itself
- **Correct: B**



Activity

- Bring 3 to 4 recent **news headlines about AI** (you can pick from health, education, ethics, etc). Example:
- “Hospitals are turning to AI for cancer diagnosis.”
- “Universities worry about ChatGPT and plagiarism.”
- “AI deepfakes raise new challenges for election security.”
- Task:
- **Group Task:** Brainstorm possible **research questions** inspired by it.
 - At least 2 bad/vague questions → **Critique:** Why are the “bad” ones weak?
 - At least 1 good/specific question → **Critique:** Why are the “good” ones weak?
- Write them down.



Checklist: Features of a Good Research Question

- **Clear & Specific**

- ❌ Not vague like *“Is AI good or bad?”*
- ✅ Instead: *“How do university students use ChatGPT for academic writing?”*

- **Neutral (No Bias / No Value-Laden Words)**

- ❌ *“Are lazy students failing because of TikTok addiction?”*
- ✅ *“What study habits are associated with lower exam performance?”*

- **Researchable (Answerable with Data)**

- Must be possible to answer through **observation, experiment, survey, or analysis.**
- ❌ *“Will AI destroy humanity?”* → not researchable.
- ✅ *“How do teachers perceive AI’s role in classroom*

assessment?”

- **Not Too Broad / Not Too Narrow**

- ❌ Too broad: *“What is the impact of AI on global education?”*
- ❌ Too narrow: *“How does one 15-year-old use Duolingo?”*
- ✅ Balanced: *“What are the challenges faced by high school students using Duolingo in language learning?”*

- **Relevant & Significant**

- Connects to a **real-world issue, academic gap, or societal concern.**
- Example: *“How do AI-based writing tools influence students’ ability to think critically?”*

